Equivalences of precategories are exactly the fully faithful and split essentially surjective functors

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:ffeso' (source lines 588-594, Lemma 9.4.5); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The two item labels item:ct:ffeso1 and item:ct:ffeso2 are defined inside the retained statement and are the two types being shown equivalent; they need no external entry.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained counter values (chapter 9, section 4, theorem offset 4) make the counted lemma print as Lemma 9.4.5, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial scope. One original result of the HoTT Book, counted once: equivalences of precategories are the fully faithful split essentially surjective functors (Lemma 9.4.5).

Allowed standard prerequisites (editorial).

- Equivalence of precategories. F is an equivalence of precategories when there is a functor G with natural isomorphisms G F = 1 and F G = 1.. It is the first of the two equivalent types.
- Fully faithful functor. F is fully faithful when each map hom_A(a,b) -> hom_B(Fa,Fb) is an equivalence.. It is one half of the second type and is used to build the candidate inverse functor.
- Split essentially surjective functor. F is split essentially surjective when for every b : B there is a specified a : A with Fa = b.. It supplies the object map of the candidate inverse without using truncation.

Lemma 9.4.5. *For any precategories A and B and functor $F : A \to B$, the following types are equivalent.*

(i) F is an equivalence of precategories.

(ii) F is fully faithful and split essentially surjective.

Proof. Suppose F is an equivalence of precategories, with G, η, ϵ specified. Then we have the function

$$\begin{aligned} \hom_B(Fa, Fb) &\to \hom_A(a, b), \\ g &\mapsto {\eta_b}^{-1} \circ G(g) \circ \eta_a. \end{aligned}$$

For $f : \hom_A(a, b)$, we have

$${\eta_b}^{-1} \circ G(F(f)) \circ \eta_a = {\eta_b}^{-1} \circ \eta_b \circ f = f$$

while for $g : \hom_B(Fa, Fb)$ we have

$$\begin{aligned} F({\eta_b}^{-1} \circ G(g) \circ \eta_a) &= F({\eta_b}^{-1}) \circ F(G(g)) \circ F(\eta_a) \\ &= \epsilon_{Fb} \circ F(G(g)) \circ F(\eta_a) \\ &= g \circ \epsilon_{Fa} \circ F(\eta_a) \\ &= g \end{aligned}$$

using naturality of ϵ, and the triangle identities twice. Thus, $F_{a,b}$ is an equivalence, so F is fully faithful. Finally, for any $b : B$, we have $Gb : A$ and $\epsilon_b : FGb \cong b$.

On the other hand, suppose F is fully faithful and split essentially surjective. Define $G_0 : B_0 \to A_0$ by sending $b : B$ to the $a : A$ given by the specified essential splitting, and write ϵ_b for the likewise specified isomorphism $FGb \cong b$.

Now for any $g : \hom_B(b, b')$, define $G(g) : \hom_A(Gb, Gb')$ to be the unique morphism such that $F(G(g)) = {(\epsilon_{b'})}^{-1} \circ g \circ \epsilon_b$ (which exists since F is fully faithful). Finally, for $a : A$ define $\eta_a : \hom_A(a, GFa)$ to be the unique morphism such that $F\eta_a = {\epsilon_{Fa}}^{-1}$. It is easy to verify that G is a functor and that (G, η, ϵ) exhibit F as an equivalence of precategories.

Now consider the composite (i)→(ii)→(i). We clearly recover the same function $G_0 : B_0 \to A_0$. For the action of G on hom-sets, we must show that for $g : \hom_B(b, b')$, $G(g)$ is the (necessarily unique) morphism such that $F(G(g)) = {(\epsilon_{b'})}^{-1} \circ g \circ \epsilon_b$. But this equation holds by the assumed naturality of ϵ. We also clearly recover ϵ, while η is uniquely characterized by $F\eta_a = {\epsilon_{Fa}}^{-1}$ (which is one of the triangle identities assumed to hold in the structure of an equivalence of precategories). Thus, this composite is equal to the identity.

Finally, consider the other composite (ii)→(i)→(ii). Since being fully faithful is a mere proposition, it suffices to observe that we recover, for each $b : B$, the same $a : A$ and isomorphism $Fa \cong b$. But this is clear, since we used this function and isomorphism to define G_0 and ϵ in (i), which in turn are precisely what we used to recover (ii) again. Thus, the composites in both directions are equal to identities, hence we have an equivalence (i) $\simeq$ (ii). □